

Asian Deans' Forum 2026

THE RISING STARS

Women in Engineering Workshop

SEP 13 - 15 | HKUST, Hong Kong

Optimizing Service Operations via LLM-Powered Multi-Agent Simulation

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PROBLEM & CHALLENGES

Services Dominate Modern Economies (e.g., China, U.S., EU, Australia ...)

- (c.f. Manufacturing) **Human-centric: Intensive Interactions**

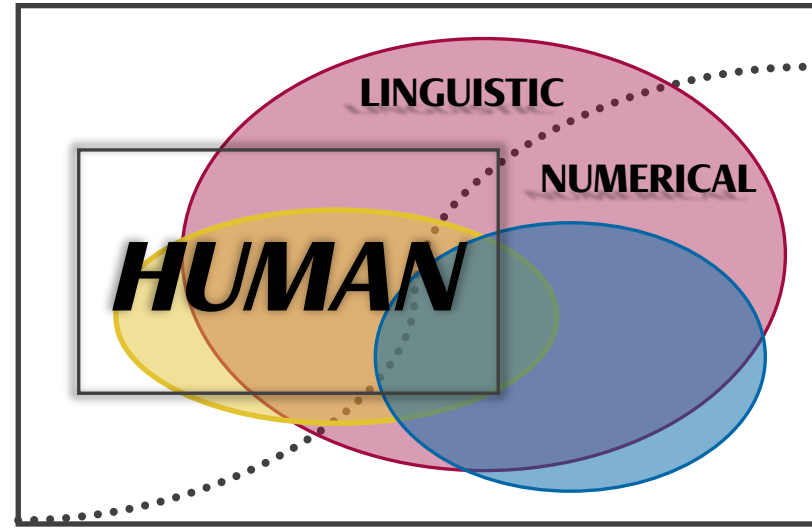
Designer's Decision-Making Problem: System Design Optimization

Systems	Supply Chains	Innovation Contests	Online Marketplaces
Designs	Sustainable Policies	Incentive Structures	Operational Levers
Participants	Multi-echelon Players	Contestants	Sellers, Consumers

Generic Task: Design Parameters → Participant Responses → Design Performance

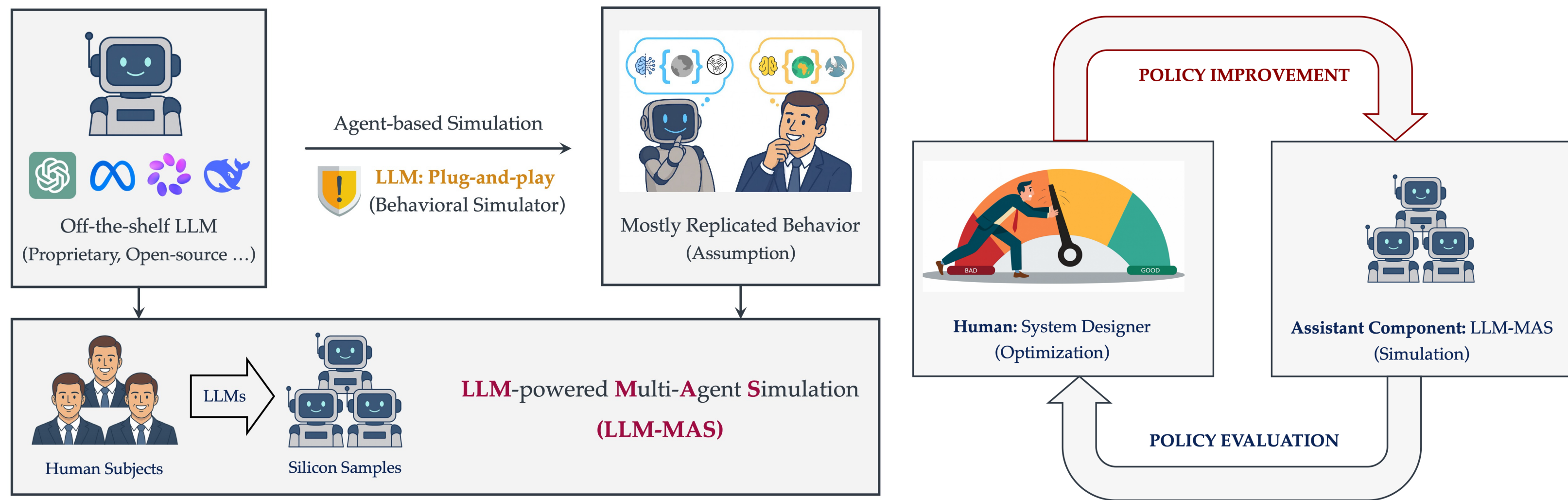
- Capturing (Human) Behavior** Is Non-trivial!

Empirical Studies	Limited Data, Insufficient Granularity
Lab Experiments	Costly, Time-consuming, Hard to Scale
Analytical Models	Idealized Scenarios, Rigid Assumptions
LLM Simulations	Promising (Academia, Industry)



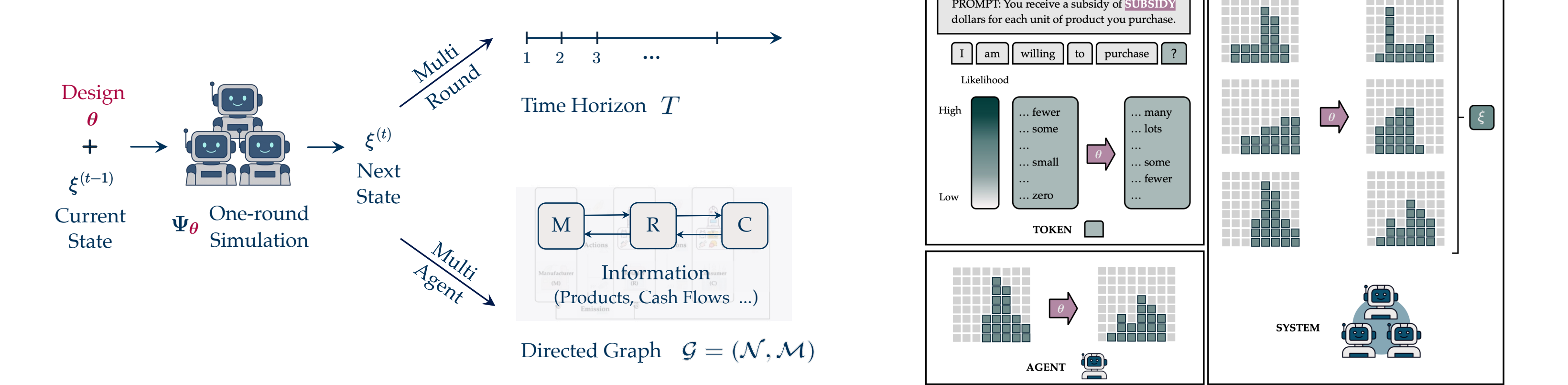
NOVEL FRAMEWORK: LLM-MAS

Our Contributions: Modeling – Algorithm – Empirics

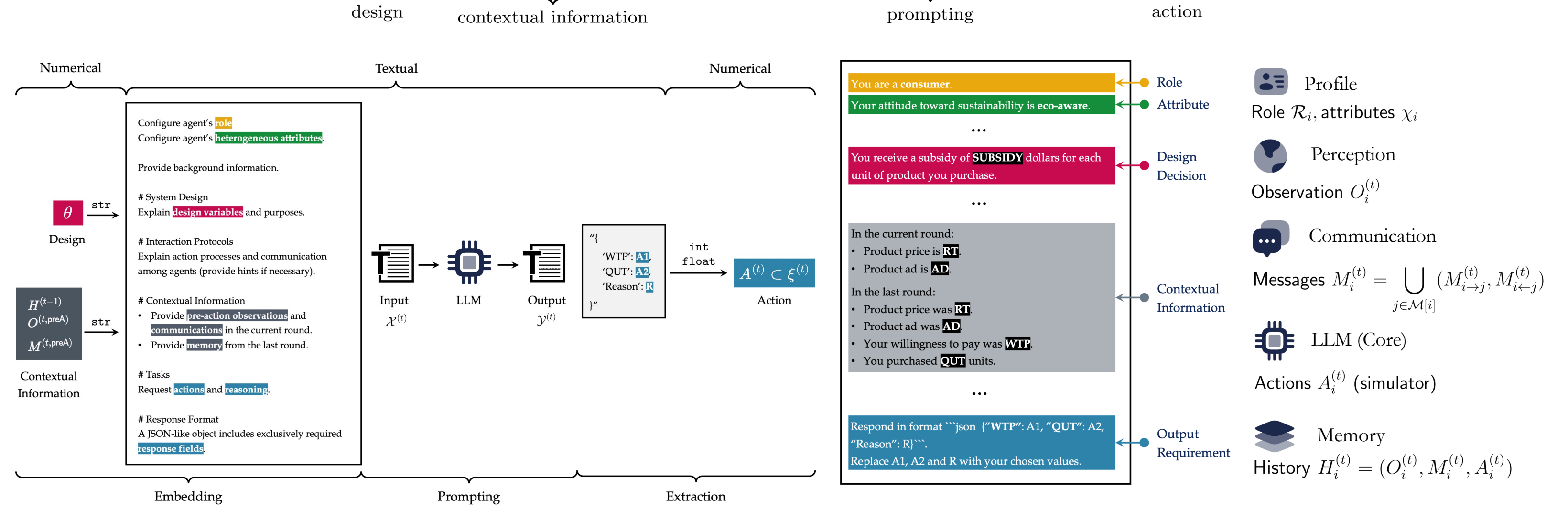


CONTRIBUTION: MODELING

Multi-Round, Multi-Agent Simulation $\langle \mathcal{G}, \Psi_{\theta}, T \rangle$



LLM Prompting



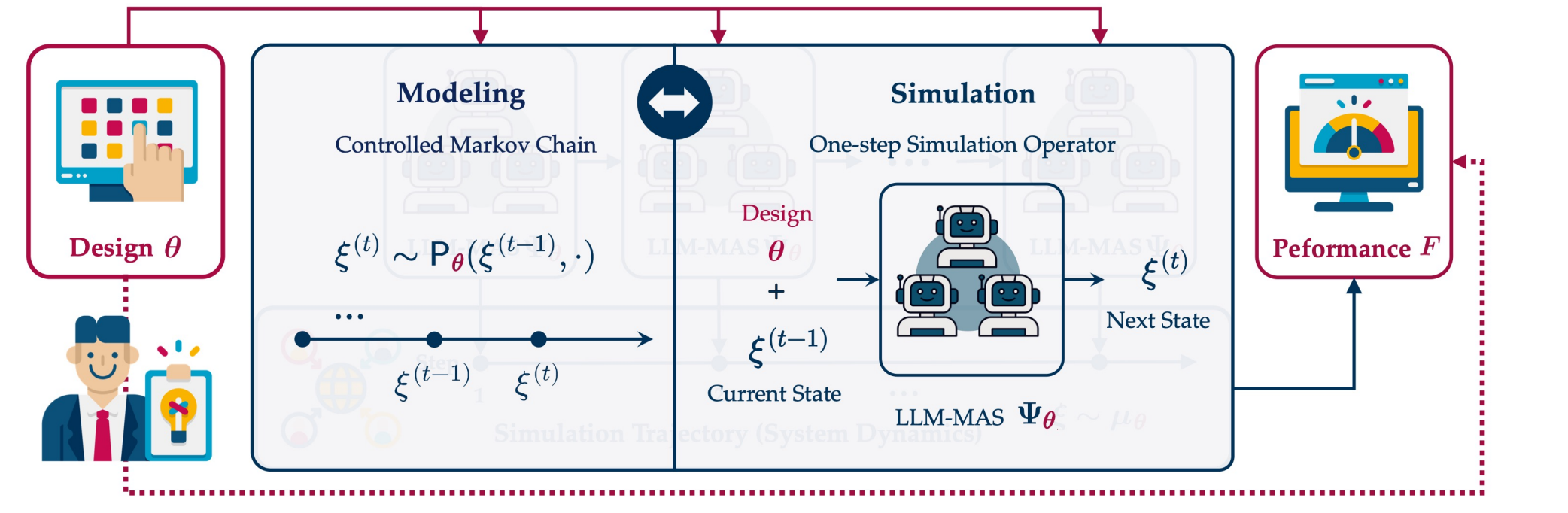
System Dynamics

- Controlled Markov Chain**

Stochastic Optimization

- Decision-dependent uncertainty**

$$\min_{\theta \in \Theta} \left\{ f(\theta) := \mathbb{E}_{\xi \sim \mu_{\theta}} [F(\theta; \xi)] \right\}$$

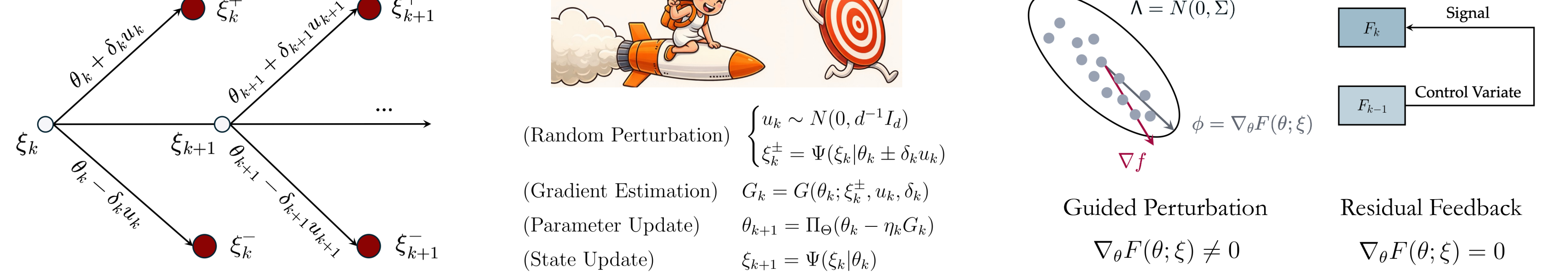


CONTRIBUTION: ALGORITHM

On-Trajectory Learning (OTL) Tackles LLM Deep Coupling, High Inference Latency

I. Zeroth-Order Approach

II. Variance Reduction



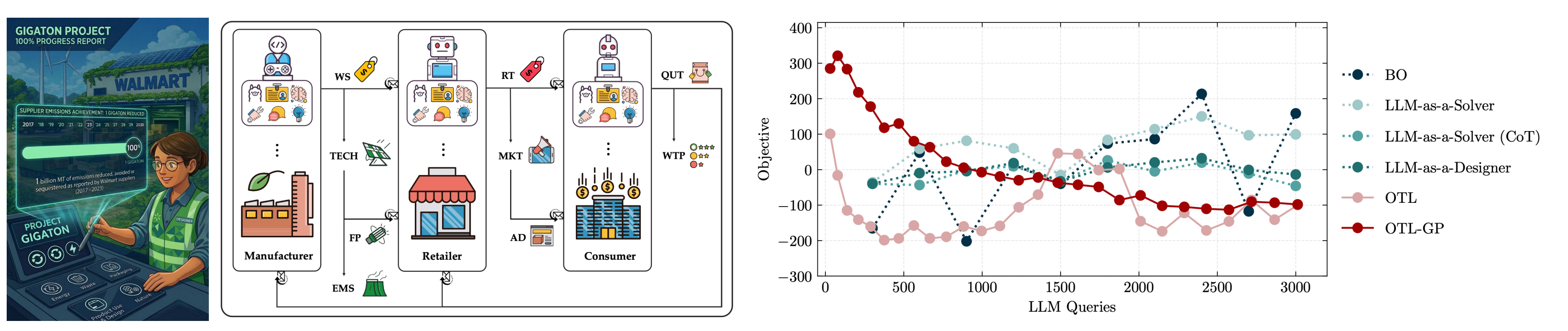
THEORETICAL GUARANTEE

Almost-Sure Convergence of OTL Iterates $\{\theta_k : k \in \mathbb{N}_+\}, \theta_{\infty} \in \Theta_0$

- Objective: Lipchitz Continuous
- Feasible Set: Compact, Convex
- Markov Kernel: Smooth, Geometric Mixing
- Transient Gradient: Bounded
- Poisson Equation Solution: Smooth
- Step-size: Two-timescale

RESULTS & IMPLICATIONS

Prototyping Example: Walmart's Project Gigaton



R1. OTL Outperforms Baselines!

R2. LLM-MAS Complements Traditional Approaches!

FUTURE RESEARCH WORKS

Potential Directions of LLM-MAS&O

- Refined Theory
- Efficient Deployment
- High-Fidelity Alignment
- Scalable Deployment
- Robust Deployment
- Boarder Applicability

Organizers

